Method of expressing concentration of solution

21. (c) $N_1V_1 = N_2V_2$, $36 \times 50 = N_2 \times 100$

$N_2 = \frac{36\times50}{100} = 18$; $18NH_2SO_4 = 9MH_2SO_4$.

22. (c) Molarity $= \frac{w}{m.wt.\times \text{volumeinlitre}} = \frac{171}{342\times1} = 0.5M$.

23. (c) $N_1V_1 + N_2V_2 = NV$

$4x + 10(1-x) = 6 \times 1$; $-6x = -4$; $x = 0.66$

24. (b) $[H_3O^+] = 2 \times 0.02 = 0.04M$

$\therefore$ 2 *litre* solution contains 0.08 mole of H_3O^+.

25. (c) $\because$10 *litre* of urea solution contains 240 *gm* of urea

$\therefore$ Active mass= $\frac{240}{60\times10} = 0.4$.

26. (d) $NV = N_1V_1 + N_2V_2 + N_3V_3$

or, $1000N = 1 \times 5 + \frac{1}{2} \times 20 + \frac{1}{3} \times 30$or $N = \frac{1}{40}$.

27. (d) $W = \frac{N\times eq.wt.\times V(ml)}{1000} = \frac{0.05\times49.04\times100}{1000} = 0.2452$.

29. (a) For HCl $M = N = 0.1$

$N_1V_1 = N_2V_2$; $25 \times N_1 = 0.1 \times 35$

$N_1 = \frac{0.1\times35}{25}$; $\therefore M = \frac{0.1\times35}{25\times2} = 0.07$.

30. (c) We know that

Molarity $= \frac{\text{Number of moles of solute}}{\text{Volume of solution in } litre}$

$\therefore 2.0 = \frac{0.5}{\text{Volume of solution in} litre}$

$\therefore$ Volume of solution in $litre$

$= \frac{0.5}{2.0} = 0.250 litre = 250 ml.$

31. (a) $M = \frac{w}{m \times V(l)}; 0.52 = \frac{w}{36.5 \times 0.15}; w = 2.84 gm$

32. (c) $M = \frac{n}{V(l)}; 0.5 = \frac{n}{2}; n = 1$

33. (d) $N = \frac{W}{M} = \frac{828}{46} = 18, n = \frac{w}{m} = \frac{36}{18} = 2$

$x_{H_2O} = \frac{n}{n+N} = \frac{2}{2+18} = \frac{2}{20} = 0.1$

34. (a) $N = \frac{w \times 1000}{E \times \text{volume in} ml.}, E = \frac{98}{3} = 32.6$

$N = \frac{4.9 \times 1000}{32.6 \times 500} = 0.3N.$

39. (b) Mole fraction of solute $= \frac{20}{80} = 0.25.$

40. (c) $N = \frac{w \times 1000}{m.wt. \times \text{Volume in } ml} = \frac{4 \times 1000}{40 \times 100} = 1.0N.$

21.